

Continuous Travel In Virtual Reality Using a 3D Portal

Andrew Atkins
Cornell Tech
USA
aja323@cornell.edu

Serge Belongie
Cornell University
USA

Harald Haraldsson
Cornell Tech
USA

ABSTRACT

In virtual reality (VR), continuous movement often leads to users experiencing vection, or cybersickness. To circumvent vection, users are typically given the choice to use teleportation. However, teleportation leads to less spatial awareness compared to continuous movement. One approach to combating cybersickness in continuous movement is to restrict the user's field of view (FOV). This is typically done by occluding the user's peripheral vision with a vignette. We developed a FOV restricting continuous locomotion system that does not occlude the user's FOV and instead uses a 3D portal to display the continuous movement in a limited area in the user's FOV. We found that this system reduces nausea and disorientation compared to continuous locomotion. However, our system did not significantly increase spatial awareness compared to teleportation.

CCS CONCEPTS

• Human-centered computing → Virtual reality.

KEYWORDS

virtual reality, travel, locomotion

ACM Reference Format:

Andrew Atkins, Serge Belongie, and Harald Haraldsson. 2021. Continuous Travel In Virtual Reality Using a 3D Portal. In *The Adjunct Publication of the 34th Annual ACM Symposium on User Interface Software and Technology (UIST '21 Adjunct)*, October 10–14, 2021, Virtual Event, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3474349.3480227>

1 INTRODUCTION

Continuous movement in VR results in increased spatial awareness compared to teleportation [1]. However, continuous movement often leads to vection, also known as cybersickness, due to a mismatch between the signals received by a user's visual senses and vestibular system [4]. Because of this, teleportation is often chosen over continuous locomotion to avoid vection. This research aims to develop and evaluate a locomotion system that causes minimal vection and results in high spatial awareness.

In the proposed locomotion system, hereby referred to as the portal technique, users travel using continuous movement, however, this continuous movement is only viewed through a head stabilized

3D window (Fig. 2). When users initiate travel, the 3D window will appear, and through this window, users will view the path of their continuous travel. However, this window does not cover the user's entire field-of-view, and therefore the periphery will remain stationary. When travel is completed, users will teleport to the travel destination and the 3D window will disappear.

The following hypotheses are expected to hold for the proposed system of travel:

- H1: The system will reduce cybersickness rates compared to joystick continuous movement due to the reduced FOV in which continuous movement occurs [5].
- H2: The continuous movement employed by the system will result in better spatial awareness in users than teleportation [1].

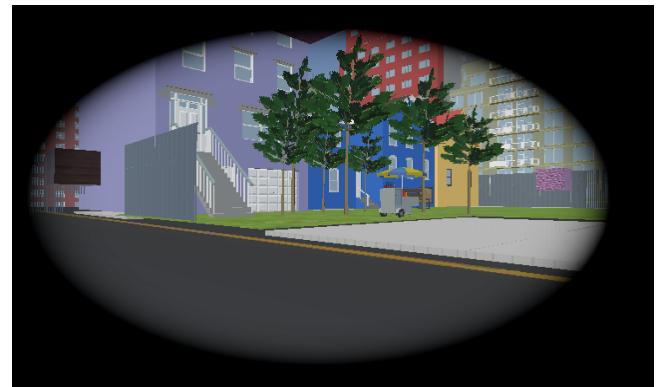


Figure 1: User's view in the restricted FOV technique.

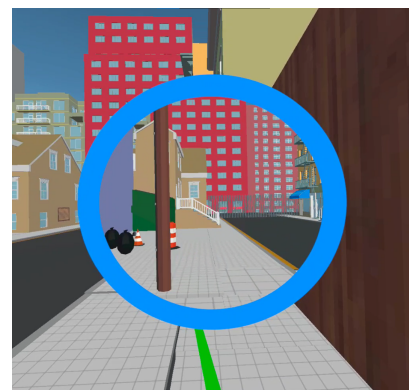


Figure 2: User's view in the portal technique.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UIST '21 Adjunct, October 10–14, 2021, Virtual Event, USA

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8655-5/21/10...\$15.00

<https://doi.org/10.1145/3474349.3480227>

2 METHOD

Four locomotion systems were tested: Joystick (JS) continuous, teleportation, restricted field-of-view (FOV), and the proposed portal locomotion system. JS continuous locomotion is a system in which users move throughout the environment in continuous increments using the joystick. Teleportation consists of instant transportation from one location in the environment to another. Restricted FOV is similar to JS continuous, however when the user begins moving a black vignette appears in their vision, thus reducing their FOV (Fig. 1). The portal technique is most like the restricted FOV technique in that the peripheral view does not display continuous optical flow. However, in the restricted FOV technique, the user's peripheral FOV is rendered black, while in our technique the peripheral view is still present but not moving.

To test users' spatial awareness, a user study similar to [2] was conducted. Users were asked to repeatedly move from one part of the environment to another in a straight line. Upon arriving at the indicated destination, a square was shown to the user, which displayed a common city object (Fig. 3). This object was located somewhere in the environment (Fig. 4). The user was asked to find this object, point to it with a raycast, and then confirm their selection with a button press. The time between when the user was shown the object and the time that they found and selected the object in the environment was recorded. Each user performed eight pointing tasks with each of the locomotion systems, for a total of 32 pointing times recorded per user.

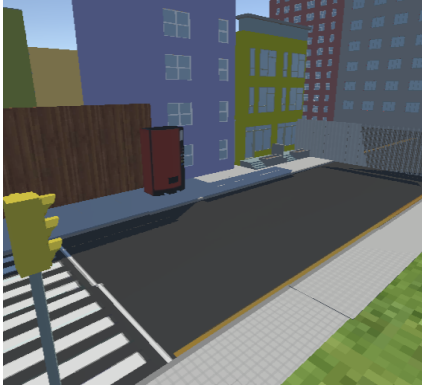


Figure 3: A birds-eye view of objects in the virtual environment.

This task was designed to test how spatially aware users were when arriving at the destination. As described in [2], if users were completely disoriented when arriving at the destination, then the time between being shown the object and finding the object in the environment should be close to a random visual search. However, if the users still had some spatial awareness, then they should be able to recall where in the environment the object was located and point to it.

Once a user completed the pointing tasks for each locomotion system, he or she was then asked to fill out a corresponding section in the accompanying survey. This survey contained a Simulator Sickness Questionnaire (SSQ) and Slater-Usch-Steed (SUS) questionnaire for each of the four locomotion techniques. Additionally,



Figure 4: The city object indicated in the users field-of-view.

Table 1: Average Pointing Times

Continuous	Portal	Teleport	FOV
2.5914	3.1408	3.1674	3.0084

Table 2: Pointing Time Significant Differences

	Continuous	Portal	Teleport	FOV
Continuous	-	Yes	Yes	No
Portal	Yes	-	No	No
Teleport	Yes	No	-	No
FOV	No	No	No	-

at the end of the survey participants were given the chance to include any additional thoughts they had.

3 RESULTS

3.1 Pointing Times

A total of ten participants (1 female, 8 male, 1 non-binary) took part in the experiment. Each participant completed the pointing tasks for all four locomotion techniques. We recorded eight pointing times for each technique, resulting in a total of 320 pointing times. The average pointing time for each technique can be found in Table 1.

A Shapiro Wilks test was conducted on the pointing latencies, which indicated that the data was not normally distributed. Therefore, we performed the Wilcoxon Signed Rank test on the pointing times for each pair of locomotion techniques ($\alpha = 0.05$).

No significant difference was found for pointing times between teleportation, restricted FOV, and the portal technique (Table 2). However, the portal and teleportation techniques were found to result in significantly slower pointing times compared to the continuous technique.

3.2 Cybersickness

The results of the SSQ were converted into four scores as described in [3]. These scores were total score (TS), nausea (N), disorientation

Table 3: Average SSQ Scores

	Continuous	Portal	Teleport	FOV
TS	23.936	13.838	5.236	21.318
N	16.218	7.632	4.77	13.356
O	17.826	12.128	4.548	19.93
D	33.408	15.312	4.176	29.232

(D), and oculomotor (O). Total score indicates the overall effects of simulator sickness, nausea indicates how sick the system made the user feel, disorientation relates to feelings of dizziness and vertigo, and oculomotor relates to effects on the user’s eyes and head. A higher score indicates greater effects of simulator sickness. The averages for each score corresponding to each technique can be found in Table 3.

Upon conducting the Shapiro Wilks test on the four SSQ scores across locomotion methods, we found that the data was not normally distributed. Thus, we conducted the Wilcoxon Signed Rank test for each pair of locomotion techniques and each of the four scores ($\alpha = 0.05$). This revealed that the portal and restricted FOV techniques did not result in significantly lower total scores compared to the continuous technique, whereas the teleport technique did.

Next, we compared the nausea, disorientation, and oculomotor scores across techniques. This revealed that, with regards to nausea and disorientation, the portal technique resulted in significantly lower scores compared to the continuous technique. Interestingly, the restricted FOV technique did not result in significantly lower scores compared to the continuous technique in any categories. Compared to the other three techniques, the teleport technique resulted in significantly lower scores in all categories.

3.3 Presence

The SUS questionnaire contained six questions relating to the users’ sense of presence. These questions were responded to on a scale from one to seven, with one indicating little presence and seven indicating high presence. Table 4 displays the average responses for each question per locomotion technique. The results were compared using T-Tests, with the following three questions resulting in significant differences:

- Q1: “Please rate your sense of being in the virtual environment.”
- Q4: “During the time of the experience, which was strongest on the whole, your sense of being in the virtual environment, or of being elsewhere?”
- Q6: “During the time of the experience, did you often think to yourself that you were actually in the virtual environment?”

The continuous technique resulted in significantly higher scores compared to the other three techniques for question one. However, for question four the portal and teleportation technique resulted in significantly lower scores compared to the continuous technique, while the restricted FOV technique did not. Finally, for question six, only the portal technique resulted in significantly lower scores compared to the continuous technique.

Table 4: Average SUS Scores

	Continuous	Portal	Teleport	FOV
Q1	5.8	4.8	4.9	4.7
Q2	5.4	4.5	5	4.7
Q3	5.8	5.2	4.6	5.2
Q4	6.1	4.4	5.7	5.4
Q5	6	5.3	5.4	5.3
Q6	5.2	4.2	4.6	4.4

4 DISCUSSION

4.1 Cybersickness

Our first hypothesis, that the portal technique would result in significantly less cybersickness than continuous locomotion was partially supported by our results. In the results of the SSQ, there was no significant difference between the continuous and portal techniques with respect to the total and oculomotor scores. However, the portal technique did result in significantly lower nausea and disorientation scores compared to the continuous technique.

Interestingly, the portal technique resulted in significantly lower scores in all categories when compared to the restricted FOV technique. Since both techniques reduce motion cues in the peripheral vision, we expected the techniques to not be significantly different. As expected, however, the teleportation technique resulted in significantly lower scores across all categories compared to all other techniques. This suggests that the portal technique could act as a good alternative to the restricted FOV technique when attempting to reduce cybersickness.

4.2 Spatial Awareness

Our second hypothesis, that the continuous movement employed by the portal technique would result in greater spatial awareness than teleportation, was not supported. The portal and teleportation techniques both resulted in significantly higher pointing times compared to the continuous technique. However, the restricted FOV technique did not result in significant differences with any of the other three techniques.

It was surprising that the portal technique resulted in significant differences compared to the continuous technique, while the restricted FOV technique did not. We suspect that this was because of differences in the size of the area in which the continuous motion took place. The restricted FOV vignette was oval shaped, whereas the portal technique had continuous motion occurring in a circular region. In future work, we plan to test different sizes for the areas of continuous movement to further compare the portal and restricted FOV techniques.

4.3 Presence

As expected, there were no significant differences between the restricted FOV and portal techniques for any of the six SUS questions. However, where the portal technique indicated significantly lower presence compared to the continuous technique, the restricted FOV technique did not. Therefore, while the portal and restricted FOV technique are similar with regards to presence, it is possible that

the differences between the two techniques may have resulted in the portal technique reducing presence.

One possible explanation for why the portal technique reduced presence is due to the outline of the portal. We chose to use a blue outline so that the border of the portal was made obvious, however, it is possible that this broke immersion. Therefore, we plan to alter the portal outline to make it stand out less within the virtual environment, hopefully leading to increased presence.

REFERENCES

- [1] Niels H. Bakker, Peter O. Passenier, and Peter J. Werkhoven. 2003. Effects of Head-Slaved Navigation and the Use of Teleports on Spatial Orientation in Virtual Environments. *Human Factors: The Journal of the Human Factors and Ergonomics Society* 45, 1 (2003), 160–169. <https://doi.org/10.1518/hfes.45.1.160.27234>
- [2] D A Bowman, D Koller, and L F Hodges. 1997. Travel in immersive virtual environments: an evaluation of viewpoint motion control techniques. In *Proceedings of IEEE 1997 Annual International Symposium on Virtual Reality*. 45–52.
- [3] Robert S. Kennedy, Norman E. Lane, Kevin S. Berbaum, and Michael G. Lilienthal. 1993. Simulator Sickness Questionnaire: An Enhanced Method for Quantifying Simulator Sickness. *The International Journal of Aviation Psychology* 3, 3 (1993), 203–220. https://doi.org/10.1207/s15327108ijap0303_3
- [4] Joseph J. Laviola. 2000. A discussion of cybersickness in virtual environments. *ACM SIGCHI Bulletin* 32, 1 (2000), 47–56. <https://doi.org/10.1145/333329.333344>
- [5] Joseph J. LaViola, Ernst Kruijff, Ryan P. McMahan, Doug A. Bowman, and Ivan Poupyrev. 2017. *3D user interfaces: theory and practice*. Addison-Wesley.